CAR-CVGL: Conditional height-Aware BEV Representation for Cross-View Geo-Localization

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Abstract

This paper addresses the problem of cross-view geo-localization, where the goal is to estimate the vehicle pose by matching ground observations to aerial imagery. Existing methods typically learn feature alignment implicitly and often discard height information during bird's-eye-view (BEV) construction, limiting robustness and introducing ambiguity. To overcome these limitations, we propose a transformer-based framework that reformulates cross-view geo-localization as an explicit conditional alignment problem in BEV space. Our method introduces a conditional height-aware feature selection mechanism, where satellite BEV features condition the selection of the most relevant height-wise ground features prior to BEV pooling, preserving informative 3D structure and reducing ambiguity. In addition, we propose a spatial cross-attention module that directly couples ground and satellite BEV features through direct cross-view feature exchange, enabling more robust alignment under challenging conditions. Experiments on FordAV and KITTI benchmarks indicate consistent improvements over state-of-the-art methods. In particular, our approach achieves superior performance under cross-area evaluation, demonstrating improved generalization across unseen geographic regions and appearance variations, while reducing localization error by 33.8% compared to the previous state of the art on FordAV. Furthermore, we introduce a lightweight variant, CAR-Lite, which achieves 8.55 FPS while maintaining competitive localization accuracy. Project page: https://npoul.github.io/car-cvgl/

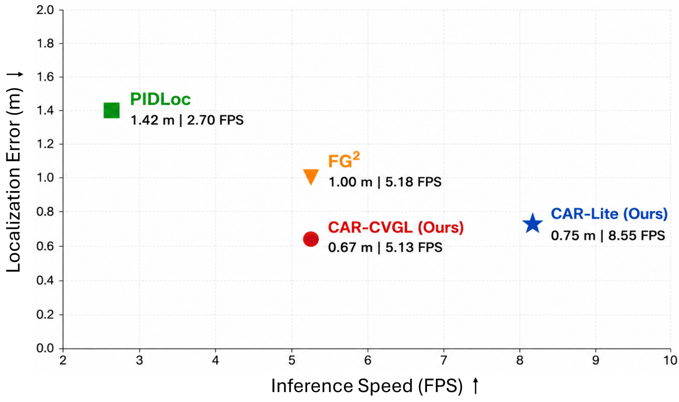


Figure 1: Inference speed vs localization error comparisons between our full and lightweight models and other state-of-the-art methods tested on an Nvidia RTX 3090 GPU.

1 Introduction

Cross-view geo-localization aims to estimate vehicle pose by matching ground observations to aerial imagery. Although the global navigation satellite system (GNSS) is commonly used for geographic positioning, it is vulnerable to interference, jamming, and signal loss in challenging environments [24, 45]. Thus, cross-view localization has attracted increasing attention due to its importance in autonomous driving [12, 31, 36, 38], UAV applications [4, 5], robot navigation [16] and GNSS-denied [16, 40] scenarios, where reliable localization is challenged by substantial viewpoint and appearance variations. Compared to traditional localization systems that rely on expensive sensors such as LiDAR or high-precision GNSS, vision-based cross-view localization offers a scalable and cost-effective alternative by leveraging widely available satellite imagery and onboard cameras [6].

Despite recent progress, precise cross-view geo-localization remains constrained by two key factors. First, cross-view feature alignment is often learned implicitly through the training objective [36], lacking an explicit mechanism to directly couple ground and aerial representations. This limitation can reduce robustness under cross-area appearance variations and viewpoint discrepancies. Second, the construction of ground features in bird's-eye-view (BEV) space typically discards information along the height dimension [30], which introduces ambiguity and weakens cross-view correspondence. Although BEV representation provides a geometrically consistent encoding space, most existing cross-view localization methods typically construct ground and aerial representations independently and perform feature alignment only at the matching stage. However, due to the severe viewpoint variations between ground and aerial observations, independently learned representations often exhibit weak geometric consistency and ambiguous cross-view correspondence.

We argue that cross-view correspondence should not be treated solely as a downstream matching problem, but should instead directly influence the construction of the ground BEV representation. Motivated by this observation, we reformulate cross-view geo-localization as an explicit conditional alignment problem in BEV space, where aerial context directly guides both feature construction and cross-view correspondence. To this end, we introduce a novel conditional height-aware feature selection mechanism, in which satellite BEV features guide the selection of the most informative height-wise ground features prior to BEV aggregation. By conditioning the ground-to-BEV transformation on cross-view cues, the proposed method preserves relevant 3D structure while reducing ambiguity in the resulting representation. Furthermore, we incorporate an explicit spatial cross-attention module that directly couples ground and satellite BEV features, establishing structured cross-view interactions for precise alignment. By allowing aerial context to directly influence ground-to-BEV feature construction, the proposed approach enables more informative and geometrically consistent feature aggregation, leading to improved cross-view correspondence and robustness under challenging viewpoint and appearance variations. Experiments demonstrate that our method achieves consistent improvements over the compared state-of-the-art methods in terms of localization accuracy, while our lightweight variant provides substantially faster inference (8.55 FPS) while maintaining competitive performance, as illustrated in Fig. 1.

We summarize our main contributions as follows:

(1) We introduce a conditional height-aware BEV construction strategy that leverages satellite BEV features to guide the aggregation of height-wise ground features, mitigating ambiguity introduced by naive pooling.

(2) We propose an explicit cross-view feature coupling mechanism based on spatial cross-attention, enabling direct information exchange between ground and satellite BEV representations for robust alignment.

(3) We develop a unified transformer-based framework that jointly models BEV construction and cross-view interaction in an end-to-end manner.

(4) We achieve state-of-the-art performance on benchmark datasets, including FordAV [1] and KITTI [8], with particularly strong improvements under cross-area evaluation, reducing localization error by up to 33.8% and demonstrating enhanced robustness and generalization under unseen conditions.

(5) We introduce a lightweight variant, namely CAR-Lite, that significantly improves efficiency while maintaining competitive localization accuracy, enabling practical deployment in real-time and resource-constrained scenarios.

2 Related Work

2.1 Bird's-Eye-View Representation

To mitigate the large perspective gap between ground-level and aerial views, recent works introduce bird's-eye-view (BEV) representations [14, 23] as an intermediate space for cross-view matching. By projecting ground features into a top-down representation, BEV-based methods aim to align spatial structures more directly with aerial imagery. Although LiDAR and radar-based systems achieve high localization accuracy [9, 15, 17], their reliance on specialized hardware limits scalability and real-world deployment. Camera-based approaches provide a more practical alternative, often leveraging multi-camera setups to construct BEV representations, primarily for perception tasks such as segmentation and scene understanding [10, 14, 34, 41]. Homography-based methods, such as Inverse Perspective Mapping (IPM), approximate BEV representations under the assumption of a planar ground surface [25]. Recent transformer-based approaches attempt to learn BEV representations adaptively using attention mechanisms. Such approaches have demonstrated strong performance by learning to construct BEV features directly from image inputs while implicitly capturing underlying 3D structure without relying on hand-crafted geometric priors [10, 14, 34, 41]. Finally, methods that model image-to-BEV transformation using probabilistic feature representations, such as Gaussian-based formulations, have recently gained increasing attention, enabling robust BEV feature construction, especially in the presence of depth ambiguity and occlusions [3, 19].

2.2 Cross-View Geo-Localization

Cross-view geo-localization (CVGL) aims to estimate geographic location by matching ground-level observations to geo-referenced aerial imagery. Early approaches predominantly adopt two-stream convolutional architectures trained to learn an embedding space where global descriptors are optimized to minimize the distance between matching ground–aerial pairs [37, 42]. While effective, these methods struggle with the large appearance and geometric discrepancies between views.

Geometry-based methods typically rely on homography to establish correspondences between ground and aerial views [13, 26, 28, 33]. However, such methods inherently rely on a flat-ground assumption and therefore fail to properly exploit above-the-ground features, especially in urban scenes. [31] aims to address this limitation by introducing a top-down feature aggregation module in order to enrich on-ground points with an above-view perspective appearance.

Transformer-based architectures [7, 30, 43] introduce global attention mechanisms to better capture long-range dependencies and improve robustness under significant viewpoint and appearance variations.

BEV transformation has been widely adopted in cross-view localization as an intermediate spatial representation for bridging the large viewpoint gap between ground-level and aerial imagery, by projecting features into a common top-down reference space [13, 35, 36, 38]. This representation reduces perspective distortion, improving robustness to viewpoint variations and partial occlusions. FG^2 [36] demonstrated the feasibility of establishing local feature correspondences between ground and aerial images by propagating BEV point matches to the image plane. However, the construction of ground BEV features discards information along the height dimension, which introduces ambiguity and degrades localization performance. To address this limitation, recent approaches aim to explicitly estimate depth by leveraging auxiliary depth predictors [35, 38] or additional sensing modalities, such as LiDAR [12, 32, 39]. However, these dependencies introduce additional computational overhead with potential error propagation, or rely on specialized high-cost hardware.

On the contrary, our method aims to deal with these limitations by reformulating cross-view geo-localization as an explicit conditional alignment problem in BEV space. Instead of relying on explicit depth supervision or additional sensing modalities, we introduce a conditional height-aware feature selection mechanism that preserves informative vertical structure during ground-to-BEV transformation, together with a spatial cross-attention module that explicitly couples ground and satellite BEV features for robust alignment.

3 Method

3.1 Overview

An overview of the proposed CAR-CVGL framework is illustrated in Fig. 2. Given a ground-view image I_G and a geo-referenced aerial image I_A, our objective is to estimate the 3-DoF

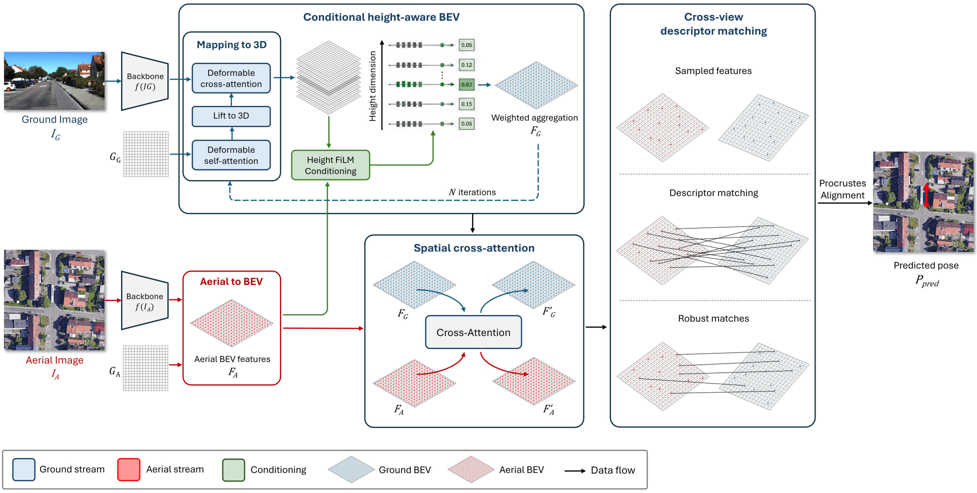


Figure 2: Overview of the proposed framework. Our method first constructs BEV representations for the aerial view image I_A and subsequently for the ground view image I_G, leveraging aerial context to guide ground-to-BEV feature aggregation through the height-aware BEV construction module. Then, a spatial cross-attention module explicitly couples ground and aerial features to establish robust correspondence between these views. Finally, point-wise descriptor matching followed by Procrustes alignment is performed to estimate the pose of the ground camera.

vehicle pose P_{pred} by establishing dense geometric correspondences between the two views in a shared bird's-eye-view (BEV) representation space.

The proposed framework consists of four main stages. First, dense visual features are extracted independently from the ground and aerial images using a pretrained image encoder. The aerial features are then projected directly into the BEV space to construct the aerial BEV representation, while the ground-view features are lifted into a discretized 3D BEV volume using deformable cross-attention and camera geometry.

Second, we introduce a conditional height-aware BEV construction module, where aerial BEV features guide the aggregation of height-wise ground features prior to BEV pooling. More specifically, aerial contextual information is used to modulate the lifted ground features through FiLM-based conditioning, enabling the network to selectively preserve informative vertical structures during the ground-to-BEV transformation process.

Third, the resulting ground and aerial BEV representations are further refined through a spatial cross-attention module that explicitly couples the two views via bidirectional feature interaction. This stage strengthens spatial correspondence and improves robustness under large viewpoint and appearance discrepancies.

Finally, dense descriptor matching is performed between the refined ground and aerial BEV representations to establish cross-view correspondences. The final vehicle pose is then estimated through weighted Procrustes alignment using the matched BEV point pairs.

3.2 Conditional height-aware BEV representation

We first extract dense feature representations $f(I_A)$ and $f(I_G)$ from the aerial and ground images respectively using a pretrained image encoder [21]. Then, the proposed BEV mapping transform produces descriptors $\mathbf{F}_A, \mathbf{F}_G \in \mathbb{R}^{H\times W\times C}$ for each BEV grid $\mathcal{G}_A$, $\mathcal{G}_G$ for the aerial and ground views respectively, where $H \times W$ represent the spatial dimensions and C the descriptor dimensionality.

Aerial-to-BEV construction: Since aerial imagery is naturally captured from a top-down perspective, its image plane is already geometrically aligned with the BEV reference frame. Therefore, the aerial BEV representation is obtained by directly projecting the aerial feature map onto the predefined BEV grid $\mathcal{G}_A$ through spatial interpolation followed by a lightweight projection head to generate the final descriptor representation:

$$\mathbf{F}_A = \mathrm{Proj}\left(\mathcal{P}(f(I_A))\right), \tag{1}$$

where $\mathcal{P}(\cdot)$ denotes the BEV projection operator, $\mathrm{Proj}(\cdot)$ the projection head and $\mathbf{F}_A \in \mathbb{R}^{H\times W\times C}$ represents the resulting aerial BEV feature map.

Ground-to-BEV 3D mapping: We follow an approach similar to [14, 36] in order to map ground image features to 3D. First, a discretized 3D BEV grid $\mathcal{V} = (x_i, y_j, z_k)$ is defined, where (x_i, y_j) correspond to the BEV plane coordinates and z_k denotes discrete height levels. Each 3D point is projected onto the image plane using the camera parameters, yielding image coordinates $\mathbf{u}_p = (u_x, u_y)$. For each projected location, deformable cross-attention [44] is employed to retrieve image features from ground-view feature map $f(I_G)$ and lift them into the BEV volume:

$$\mathbf{f}_G^{(i,j,k)} = \sum_{l=1}^{L} W_l \left(\sum_{m=1}^{M} A_{l,m} \cdot f(I_G)_{\mathbf{u}+\Delta\mathbf{u}_{l,m}} \right), \tag{2}$$

where L denotes the number of attention heads and W_l the corresponding attention weights, M denotes the number of sampling points per head, $A_{l,m}$ the learned attention weights generated from learnable queries $\mathbf{q}^{(i,j)}$ and $\Delta\mathbf{u}_{l,m}$ the learned spatial offsets.

Prior to the 3D feature lifting stage, each learnable BEV query $\mathbf{q}^{(i,j)}$ is refined through a deformable self-attention layer [44] applied in the 2D BEV space.

FiLM conditioning: In order to explicitly incorporate aerial contextual information into the ground-to-BEV transformation process we adopt the FiLM (Feature-wise Linear Modulation) [22] method. This method performs conditional feature modulation through feature-wise affine transformations, where scaling and bias parameters are dynamically predicted from an external conditioning signal. In our framework, this signal is derived from the satellite BEV features $\mathbf{F}_A$, enabling aerial contextual cues to directly influence the modulation and aggregation of height-wise ground BEV features prior to BEV pooling.

Since the ground and aerial BEV representations are not spatially aligned due to localization ambiguity, the conditioning feature $\mathbf{c}^{(i,j)}$ is computed from $\mathbf{F}_A$ through local spatial aggregation over a neighborhood corresponding to the expected uncertainty region. This regional aggregation allows the conditioning process to leverage broader spatial context and improves tolerance to initial cross-view misalignment.

The FiLM conditioning module then predicts the feature-wise modulation parameters:

$$\gamma^{(i,j)}, \beta^{(i,j)} = \Psi\left(\mathbf{c}^{(i,j)}\right), \tag{3}$$

where $\Psi(\cdot)$ denotes a lightweight multilayer perceptron.

The modulation parameters are then applied to the ground features through a feature-wise affine transformation:

$$\tilde{\mathbf{f}}_G^{(i,j,k)} = \gamma^{(i,j)} \odot \mathbf{f}_G^{(i,j,k)} + \beta^{(i,j)}, \tag{4}$$

where $\odot$ denotes element-wise multiplication.

Conditional BEV Aggregation: The conditioned features are subsequently used to estimate height-wise selection scores, which determine the contribution of each vertical hypothesis during BEV aggregation. To this end, we estimate a height selection score for each vertical hypothesis:

$$s^{(i,j,k)} = \Phi\left(\tilde{\mathbf{f}}_G^{(i,j,k)}\right), \tag{5}$$

where $\Phi(\cdot)$ is a two-layer MLP that maps the modulated feature to a height score. The scores are normalized along the height dimension using a softmax operation and finally, the ground BEV representation is obtained through weighted height aggregation:

$$\mathbf{d}_G^{(i,j)} = \sum_{k=1}^{Z} s^{(i,j,k)} \mathbf{f}_G^{(i,j,k)}, \tag{6}$$

In practice, similarly to [14, 36], the proposed ground-to-BEV mapping transform is integrated into a unified iterative refinement framework with N iterations. At each iteration, the refined BEV representations $\mathbf{d}_G^{(i,j)}$ are reused as query embeddings for the subsequent refinement stage. Finally, the resulting BEV features are obtained after applying a lightweight projection head to generate the final descriptor representation:

$$\mathbf{F}_G^{(i,j)} = \mathrm{Proj}\left(\mathbf{d}_G^{(i,j)}\right), \tag{7}$$

3.3 Spatial cross-attention

The spatial cross-attention module aims to explicitly couple ground and aerial BEV representations through direct cross-view feature interaction. Given the aerial and ground BEV feature maps $\mathbf{F}_A, \mathbf{F}_G \in \mathbb{R}^{H \times W \times C}$ obtained from Eq. (1) and Eq. (7) respectively, the module jointly refines the two representations by enabling each view to attend to complementary features from the other view. More specifically, the ground BEV features are first projected into query embeddings, while the aerial BEV features are projected into keys and values:

$$\mathbf{Q}_G = W_Q \mathbf{F}_G, \quad \mathbf{K}_A = W_K \mathbf{F}_A, \quad \mathbf{V}_A = W_V \mathbf{F}_A, \tag{8}$$

where W_Q, W_K, and W_V denote learnable linear projection matrices.

The updated ground BEV representation is then obtained using scaled dot-product attention [29]:

$$\hat{\mathbf{F}}_G = \mathrm{Softmax}\left(\frac{\mathbf{Q}_G \mathbf{K}_A^\top}{\sqrt{d}}\right) \mathbf{V}_A, \tag{9}$$

where d denotes the feature dimensionality used for attention normalization.

Similarly, the aerial BEV representation is refined by attending to the ground BEV features:

$$\mathbf{Q}_A = W_Q \mathbf{F}_A, \quad \mathbf{K}_G = W_K \mathbf{F}_G, \quad \mathbf{V}_G = W_V \mathbf{F}_G, \tag{10}$$

$$\hat{\mathbf{F}}_A = \mathrm{Softmax}\left(\frac{\mathbf{Q}_A \mathbf{K}_G^\top}{\sqrt{d}}\right) \mathbf{V}_G. \tag{11}$$

3.4 Pose Estimation

Following [36], the final vehicle pose is estimated through dense cross-view descriptor matching followed by weighted rigid alignment. Given the refined ground and aerial BEV descriptor maps $\hat{\mathbf{F}}_G$ and $\hat{\mathbf{F}}_A$, a dense matching score matrix is first computed using pairwise descriptor similarity. A set of cross-view correspondences is then sampled according to the matching confidence scores, producing matched ground and aerial BEV point sets:

$$\mathcal{C} = \{(\mathbf{p}_G^n, \mathbf{p}_A^n, w_n)\}_{n=1}^{N_c},$$

where $\mathbf{p}_G^n$ and $\mathbf{p}_A^n$ denote the matched BEV coordinates, w_n corresponds to the associated matching confidence and N_c is the number of valid correspondences.

The final rigid transformation between the two BEV representations is estimated using the orthogonal Procrustes alignment [11]:

$$\{\mathbf{R}, \mathbf{t}\} = \arg\min_{\mathbf{R},\mathbf{t}} \sum_{n=1}^{N_c} w_n \, \|\mathbf{R}\mathbf{p}_G^n + \mathbf{t} - \mathbf{p}_A^n\|_2^2, \tag{12}$$

where $\mathbf{R}$ and $\mathbf{t}$ correspond to the estimated rotation and translation parameters, respectively. The final vehicle pose is directly recovered from the estimated rigid transformation.

3.5 Loss functions

Our framework is trained end-to-end using a convex combination of two losses to provide supervision on both the estimated pose and the cross-view correspondences, as follows:

$$\mathcal{L} = \mathcal{L}_{\text{VCE}} + \beta \mathcal{L}_M, \tag{13}$$

where $\mathcal{L}_{\text{VCE}}$ denotes the Virtual Correspondence Error (VCE) loss [2], $\mathcal{L}_M$ the descriptor matching loss [36], and β is a balancing coefficient.

Virtual Correspondence Error (VCE) loss: Aims to provide direct supervision on the estimated camera pose. Specifically, a set of N_v virtual points $\mathcal{P}_v$ is uniformly sampled in a 2D BEV metric space. Both the predicted rigid transformation $(\mathbf{R}, \mathbf{t})$ and the ground-truth transformation $(\mathbf{R}_{gt}, \mathbf{t}_{gt})$ are applied to the virtual points, and the VCE loss minimizes the mean Euclidean distance between the transformed point sets:

$$\mathcal{L}_{\text{VCE}} = \frac{1}{N_v} \sum_{n=1}^{N_v} \left\| \mathbf{R}\mathbf{p}_v^n + \mathbf{t} - (\mathbf{R}_{gt}\mathbf{p}_v^n + \mathbf{t}_{gt}) \right\|_2. \tag{14}$$

Matching loss: In addition to pose supervision, descriptor-level supervision is adopted from [36] through a dense matching loss. Since only a subset of BEV locations correspond across the two views, the loss is computed over N_s sampled correspondences. For each matched ground descriptor, the corresponding ground-truth aerial location is determined using the ground-truth pose transformation, and an InfoNCE objective [20] is applied:

$$\mathcal{L}_{\text{InfoNCE}}^G = -\log \frac{\sum_{n_G=1}^{N_s} e^{c_{n_G,\hat{n}_G}}}{\sum_{n_G=1}^{N_s} \sum_{n_A=1}^{N} e^{c_{n_G,n_A}}}, \tag{15}$$

where c denotes the descriptor similarity score.

Similarly, an aerial-to-ground InfoNCE loss is computed:

$$\mathcal{L}_{\text{InfoNCE}}^{A} = -\log \frac{\sum_{n_A=1}^{N_s} e^{c_{n_A,\hat{n}_A}}}{\sum_{n_A=1}^{N_s} \sum_{n_G=1}^{N} e^{c_{n_G,n_A}}}. \tag{16}$$

The final matching loss is obtained by averaging the two directional losses:

$$\mathcal{L}_M = \frac{\mathcal{L}_{\text{InfoNCE}}^{G} + \mathcal{L}_{\text{InfoNCE}}^{A}}{2}. \tag{17}$$

4 Experiments

We evaluate the proposed framework on standard cross-view geo-localization benchmarks and compare it against recent state-of-the-art approaches under both same-area and cross-area evaluation settings. In addition to quantitative comparisons, we provide qualitative visualizations to analyze the proposed cross-view alignment mechanism. Moreover, extensive ablation studies are performed to evaluate the contribution of each component of the framework to the overall accuracy. Finally, we investigate the computational complexity of the proposed approach by analyzing the trade-off between localization accuracy and inference speed, highlighting the efficiency of both the full CAR-CVGL model and the lightweight CAR-Lite variant.

4.1 Datasets and Evaluation Metrics

FordAV Dataset [1] contains synchronized ground-level and aerial imagery collected in urban driving environments. The dataset provides accurate pose annotations and exhibits significant viewpoint, illumination, and appearance variations, making it well suited for evaluating fine-grained cross-view localization performance. Following prior works, we adopt the commonly used training and testing splits for evaluation. Specifically, training was conducted using only log 4 data, while evaluation was performed on log 4 data for different weather conditions (same area) and log 5 data for different areas and weather conditions (cross area).

KITTI Dataset [8] contains urban driving sequences captured from a vehicle-mounted platform in Germany together with geo-referenced aerial imagery. The dataset presents substantial challenges due to viewpoint changes, dynamic objects, and complex urban structures. To further assess robustness and generalization, we perform same and cross-area evaluations using common training and testing splits.

Evaluation Metrics: Following previous works, we report mean and median localization and orientation errors. Moreover, we further decompose localization errors into longitudinal and lateral components based on the driving direction, and report the percentage of samples within given error thresholds. In all experiments, the initial pose error was set to $\pm 20\,\text{m}$ for position and $\pm 10^\circ$ for orientation.

4.2 Implementation details

The BEV grid is defined using a spatial resolution of 21×41 points covering a metric area of $49\,\text{m} \times 99\,\text{m}$ in front of the ground camera. Each ground BEV location is further lifted into 11 discretized height levels uniformly distributed within the range $[-25\,\text{m}, 25\,\text{m}]$, assuming

	Method	↓ Loc. (m)		↑ Lateral (%)		↑ Long. (%)		↓ Orien. (°)		↑ Orien. (%)	
		Mean	Median	R@1m	R@5m	R@1m	R@5m	Mean	Median	R@1°	R@5°
Same-area	HighlyAccurate [26]	5.12	4.18	20.20	72.96	25.20	82.49	5.03	3.91	14.48	59.60
	PureACL† [30]	5.45	4.68	16.74	70.70	20.57	80.61	4.88	4.64	11.49	53.80
	PIDLoc [12]	1.42	0.88	82.37	97.82	74.06	96.81	0.86	0.80	65.77	90.16
	$FG^{2\dagger}$ [36]	1.00	0.92	88.11	**100.00**	68.68	**100.00**	0.52	0.39	87.05	99.91
	CAR-Lite (Ours)	0.75	0.64	93.22	**100.00**	84.95	**100.00**	0.66	0.50	83.55	99.44
	CAR-CVGL (Ours)	**0.67**	**0.52**	**93.38**	**100.00**	**88.05**	**100.00**	**0.45**	**0.34**	**90.88**	**99.96**
Cross-area	HighlyAccurate [26]	6.58	5.62	16.35	74.33	11.30	65.03	6.04	4.99	11.30	50.00
	PureACL† [30]	6.50	6.33	14.64	64.86	13.66	66.49	5.35	4.97	10.78	50.32
	PIDLoc [12]	4.81	2.95	41.92	78.69	29.03	86.06	2.49	1.68	41.17	80.21
	$FG^{2\dagger}$ [36]	1.30	1.16	77.56	**100.00**	57.76	99.97	0.92	0.55	72.47	98.01
	CAR-Lite (Ours)	0.99	0.86	84.38	**100.00**	76.07	99.96	0.95	0.65	67.99	**98.78**
	CAR-CVGL (Ours)	**0.86**	**0.68**	**87.46**	**100.00**	**81.04**	**100.00**	**0.89**	**0.48**	**76.29**	97.48

Table 1: Comparison results on the FordAV dataset under same-area and cross-area settings. † indicates reproduced results using the official implementation under our experimental setting.

the camera center is located at 0 m. We employ a fixed number of iterative BEV refinement stages, with $N = 6$ for CAR-CVGL and $N = 1$ for CAR-Lite. For all experiments, we adopt the AdamW optimizer [18] with an initial learning rate of 1×10^{-4}. The network is trained on a single NVIDIA RTX 3090 GPU for 30 and 50 epochs on FordAV and KITTI datasets respectively using a batch size of 8.

4.3 Quantitative results

Tables 1 and 2 present the comparison results against the state-of-the-art on the FordAV and KITTI datasets respectively, under both same-area and cross-area evaluation settings. The methods marked with † correspond to the reproduced results obtained using the official implementations under our experimental setting for fair comparison. In addition to the full CAR-CVGL framework, we also evaluate a lightweight variant, named CAR-Lite, where the iterative BEV refinement process is performed using only a single iteration ($N = 1$). Although CAR-Lite sacrifices a small margin of localization accuracy compared to the full model, it remains significantly more computationally efficient while still outperforming most of the state-of-the-art methods under both evaluation settings.

FordAV: Under the same-area setting, CAR-CVGL achieves the best overall localization accuracy, reducing the mean localization error by 33.0%. In addition, CAR-CVGL achieves the lowest orientation estimation error with a mean angular error of 0.45°. The proposed approach demonstrates also superior generalization under the more challenging cross-area evaluation setting. CAR-CVGL reduces the mean localization error compared to FG^2, corresponding to a relative improvement of approximately 33.8%. Similarly, improvements are observed in both lateral and longitudinal localization errors. In particular, the substantial improvement along the longitudinal axis highlights the effectiveness of the proposed conditional height-aware BEV construction in preserving depth-related geometric information and improving spatial consistency between ground and aerial representations. Despite the reduced accuracy, CAR-Lite still outperforms most of the methods while offering significantly improved computational efficiency.

KITTI: Under the same-area scenario, CAR-CVGL achieves the best overall local-

Methods	Same-area				Cross-area			
	↓ Localization (m)		↓ Orientation (°)		↓ Localization (m)		↓ Orientation (°)	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median
CCVPE [37]	1.22	0.62	0.67	0.54	9.16	**3.33**	**1.55**	**0.84**
HC-Net [33]	0.80	0.50	**0.45**	0.33	8.47	4.57	3.22	1.63
DenseFlow [28]	1.48	**0.47**	0.49	**0.30**	7.97	3.52	2.17	1.21
FG^2 [36]	**0.75**	0.52	1.28	0.74	7.45	4.03	3.33	1.88
CAR-Lite (Ours)	0.82	0.56	1.02	0.70	**6.85**	3.67	3.15	1.99
CAR-CVGL (Ours)	**0.75**	0.54	1.01	0.67	**6.85**	3.65	2.93	1.82

Table 2: Comparison results on the KITTI dataset under same-area and cross-area settings.

ization accuracy, matching the current state-of-the-art FG^2, with a mean localization error of 0.75 m. In the more challenging cross-area setting, both CAR-CVGL and CAR-Lite demonstrate improved robustness and generalization under unseen environments, reducing the mean localization error from 7.45 m to 6.85 m compared to FG^2. Moreover, CAR-CVGL provides improved orientation estimation compared to CAR-Lite.

Although the proposed framework achieves superior localization accuracy, the orientation error is slightly higher than CCVPE, HC-Net, and DenseFlow. This can be attributed to the strong directional bias of the KITTI dataset, where most driving trajectories align with the road direction. While previous methods may implicitly exploit this global scene bias, our framework estimates pose through explicit cross-view geometric matching and BEV alignment, making orientation estimation more dependent on accurate spatial correspondences.

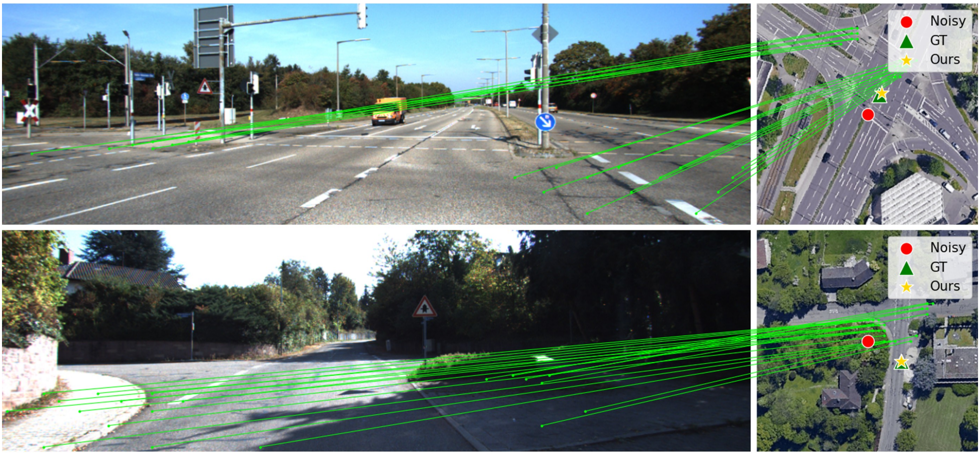
Finally, although CAR-Lite introduces a lightweight and computationally more efficient variant of the proposed framework, it still achieves competitive localization performance while maintaining lower computational complexity compared to the full CAR-CVGL model. Overall, our method is particularly effective in urban and suburban scenes, where stable vertical structures provide informative height cues. The more moderate gains on KITTI reflect its greater scene diversity and larger domain shift compared to FordAV. Experimental results on both FordAV and KITTI datasets demonstrate that the proposed mechanisms improve BEV consistency, robustness, and cross-area generalization.

4.4 Qualitative results

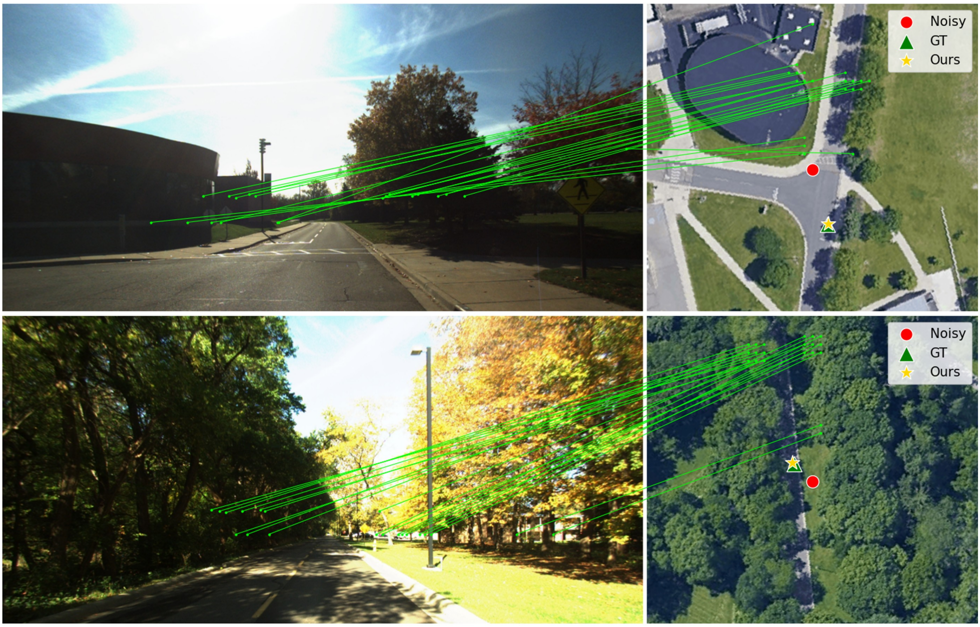
Fig. 3 presents qualitative localization results on FordAV and KITTI datasets under challenging driving scenarios. The left side of each example shows the ground-view image together with the established cross-view feature correspondences, while the right side illustrates the corresponding aerial image with the noisy initialization, ground-truth pose, and the predicted localization result obtained by the proposed framework.

The proposed framework establishes explicit dense correspondences directly in the shared BEV representation space. In addition, the proposed conditional height-aware aggregation mechanism enables the framework to preserve and selectively aggregate informative vertical structures during the ground-to-BEV transformation process, allowing the contribution of different height hypotheses to be explicitly analyzed.

The presented examples demonstrate that the proposed CAR-CVGL framework is capable of establishing geometrically consistent cross-view correspondences despite substantial



(a)



(b)

Figure 3: Qualitative localization results on the KITTI (a) and FordAV (b) datasets illustrating the established cross-view feature correspondences between ground-view and aerial images. The noisy initialization, ground-truth pose and the predicted localization result of the proposed framework are indicated by the red circle, green triangle, and yellow star, respectively.

Model	↓ Translation		↓ Lateral		↓ Longitudinal		↓ Yaw	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median
Baseline	1.00	0.92	0.52	0.41	0.78	0.68	0.52	0.39
Base + Cross	0.87	0.78	0.51	0.41	0.60	0.48	0.50	0.38
Base + Cross + SatCond	0.80	0.69	0.50	0.36	0.53	0.47	**0.44**	**0.34**
Base + Cross + SatCond + FiLM	**0.67**	**0.52**	**0.37**	**0.26**	**0.48**	**0.34**	0.45	**0.34**

Table 3: Ablation study of the proposed framework components on the FordAV dataset under the same-area setting.

viewpoint differences between ground and aerial observations. In particular, the proposed conditional BEV construction mechanism enables the network to preserve informative structural features, which remain consistent across views and facilitate robust localization.

Overall, these visualizations qualitatively validate the effectiveness of the proposed conditional cross-view alignment strategy and demonstrate the ability of the framework to achieve accurate and robust geo-localization under realistic driving conditions. Additional qualitative visualizations are also provided in the supplementary material.

4.5 Ablation study

An extensive ablation study is conducted to evaluate the contribution of each proposed component to the overall localization performance. Table 3 summarizes the results on the FordAV dataset under the same-area evaluation setting. Starting from the baseline model, we progressively incorporate the proposed spatial cross-attention module (*Cross*), the satellite-conditioned height-aware aggregation mechanism (*SatCond*), and the FiLM-based modulation module (*FiLM*) in order to analyze their individual impact. It should be noted that the *SatCond* variant performs conditioning by directly concatenating the conditioning features $\mathbf{c}^{(i,j)}$ with the ground features $\mathbf{f}_G^{(i,j,k)}$ (Eq. (2)), without applying the proposed FiLM-based feature modulation.

Introducing the spatial cross-attention module significantly improves localization performance, reducing the mean translation error by 13%, from 1.00 m to 0.87 m. The largest improvement is observed along the longitudinal axis, indicating that explicit cross-view interaction improves spatial alignment between ground and aerial BEV representations. Moreover, the satellite-conditioned aggregation further improves localization and orientation estimation accuracy, reducing the mean translation error by 8% while achieving the lowest yaw estimation error of 0.44°. Finally, incorporating the proposed FiLM-based modulation mechanism leads to the best overall performance, reducing the mean translation error by an extra 16.3% to 0.67 m and significantly improving both lateral and longitudinal localization accuracy. In particular, the substantial reduction in longitudinal error highlights the effectiveness of proposed feature modulation in producing more discriminative and geometrically consistent BEV representations.

We additionally observe that reducing the number of iterative BEV refinement stages from $N = 6$ to $N = 1$ results in the lightweight CAR-Lite variant, which improves computational efficiency by 40% (Table 4) while introducing only a moderate localization accuracy degradation of 10.7% (Table 1). This demonstrates that the proposed framework offers a flexible trade-off between computational efficiency and localization performance, while also providing an efficient solution for deployment on resource-constrained platforms.

Method	Inference Time (ms) ↓	Speed (FPS) ↑
Highly Accurate [26]	500	2.00
PureACL [30]	403	2.48
GGCVT [27]	280	3.57
PIDLoc [12]	370	2.70
FG^2 [36]	193	5.18
CAR-Lite (Ours)	**117**	**8.55**
CAR-CVGL (Ours)	195	5.13

Table 4: Inference runtime comparison. CAR-Lite achieves the fastest inference speed while maintaining competitive localization performance.

4.6 Computational Complexity

We further analyze the computational efficiency of the proposed framework and compare it against recent cross-view localization methods. Table 4 summarizes the inference time and speed measured on a single NVIDIA RTX 3090 GPU.

Although the proposed conditional aggregation and cross-attention modules introduce additional computation compared to the baseline FG^2 framework, the lightweight CAR-Lite variant significantly reduces computational cost while maintaining competitive localization accuracy. In particular, reducing the iterative BEV refinement stages from $N = 6$ to $N = 1$ decreases inference time by 40% while introducing only a moderate localization degradation, demonstrating a favorable trade-off between efficiency and localization performance.

These results highlight that the proposed framework offers a flexible balance between computational efficiency and localization accuracy, making it suitable for both high-precision and real-time deployment scenarios.

5 Conclusion

We introduce CAR-CVGL, a transformer-based framework for cross-view geo-localization that reformulates cross-view localization as an explicit conditional alignment problem in BEV space. Unlike existing approaches that construct ground and aerial BEV representations independently, the proposed framework incorporates aerial contextual information directly into the ground-to-BEV transformation process through a conditional height-aware aggregation mechanism. By leveraging aerial BEV features to guide the selection and aggregation of height-wise ground features prior to BEV pooling, the proposed approach preserves informative 3D structure and reduces ambiguity in the resulting BEV representation. In addition, we propose a spatial cross-attention module that explicitly couples ground and aerial BEV features through bidirectional feature interaction, improving spatial correspondence and robustness under large viewpoint and appearance variations. Extensive experiments on FordAV and KITTI benchmarks demonstrated state-of-the-art localization performance, particularly under the challenging cross-area evaluation setting where robust generalization is essential. We finally introduce CAR-Lite, a lightweight variant that provides a flexible trade-off between efficiency and localization performance, making it particularly suitable for real-time and resource-constrained autonomous driving applications. Current limitations include sensitivity to large initial pose errors, severe dynamic occlusions, and repetitive urban structures.

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References

[1] Siddharth Agarwal, Ankit Vora, Gaurav Pandey, Wayne Williams, Helen Kourous, and James McBride. Ford multi-av seasonal dataset. *The International Journal of Robotics Research*, 39(12):1367–1376, 2020.

[2] Eduardo Arnold, Jamie Wynn, Sara Vicente, Guillermo Garcia-Hernando, Aron Monszpart, Victor Prisacariu, Daniyar Turmukhambetov, and Eric Brachmann. Map-free visual relocalization: Metric pose relative to a single image. In *European Conference on Computer Vision (ECCV)*, pages 690–708, 2022.

[3] Florian Chabot, Nicolas Granger, and Guillaume Lapouge. GaussianBEV: 3D Gaussian Representation Meets Perception Models for BEV Segmentation. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 2250–2259, 2025.

[4] Quan Chen, Tingyu Wang, Rongfeng Lu, Yu Liu, Bolun Zheng, and Zhedong Zheng. Scale-adaptive uav geo-localization via height-aware partition learning. *arXiv preprint arXiv:2412.11535*, 2024.

[5] Quan Chen, Tingyu Wang, Zihao Yang, Haoran Li, Rongfeng Lu, Yaoqi Sun, Bolun Zheng, and Chenggang Yan. Sdpl: Shifting-dense partition learning for uav-view geo-localization. *IEEE Transactions on Circuits and Systems for Video Technology*, 34: 11810–11824, 2024.

[6] Abhilash Durgam, Sidike Paheding, Vikas Dhiman, and Vijay Devabhaktuni. Cross-view geo-localization: a survey. *IEEE Access*, 12:192028–192050, 2024.

[7] Florian Fervers, Sebastian Bullinger, Christoph Bodensteiner, Michael Arens, and Rainer Stiefelhagen. Uncertainty-Aware Vision-Based Metric Cross-View Geolocalization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 21621–21631, 2023.

[8] Andreas Geiger, Philip Lenz, Christoph Stiller, and Raquel Urtasun. Vision meets robotics: The kitti dataset. *The international journal of robotics research*, 32(11): 1231–1237, 2013.

[9] Adam W Harley, Zhaoyuan Fang, Jie Li, Rares Ambrus, and Katerina Fragkiadaki. Simple-bev: What really matters for multi-sensor bev perception? In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 2759–2765, 2023.

[10] Junjie Huang and Guan Huang. BEVDet4D: Exploit Temporal Cues in Multi-Camera 3D Object Detection. *arXiv preprint arXiv:2203.17054*, 2022.

[11] Wolfgang Kabsch. A solution for the best rotation to relate two sets of vectors. *Acta Crystallographica Section A*, 32(5):922–923, 1976.

[12] Wooju Lee, Juhye Park, Dasol Hong, Changki Sung, Youngwoo Seo, Dongwan Kang, and Hyun Myung. Pidloc: Cross-view pose optimization network inspired by pid controllers. In *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, pages 21981–21990, 2025.

[13] Ted Lentsch, Zimin Xia, Holger Caesar, and Julian FP Kooij. SliceMatch: Geometry-Guided Aggregation for Cross-View Pose Estimation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 17225–17234, 2023.

[14] Z Li, W Wang, H Li, E Xie, C Sima, T Lu, Q Yu, and J Dai. BEVFormer: Learning Bird's-Eye-View Representation from Multi-Camera Images via Spatiotemporal Transformers. *arXiv preprint arXiv:2203.17270*, 2022.

[15] Zhiwei Lin, Zhe Liu, Zhongyu Xia, Xinhao Wang, Yongtao Wang, Shengxiang Qi, Yang Dong, Nan Dong, Le Zhang, and Ce Zhu. RCEBDet: Radar-Camera Fusion in Bird's Eye View for 3D Object Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 14928–14937, 2024.

[16] Kejia Liu, Haoyang Zhou, Ruoyu Xu, Peicheng Wang, Mingli Song, and Haofei Zhang. Beyond matching to tiles: Bridging unaligned aerial and satellite views for vision-only uav navigation. *arXiv preprint arXiv:2603.22153*, 2026.

[17] Zhe Liu, Jinghua Hou, Xiaoqing Ye, Tong Wang, Jingdong Wang, and Xiang Bai. SEED: A Simple and Effective 3D DETR in Point Clouds. In *European Conference on Computer Vision (ECCV)*, pages 110–126, 2024.

[18] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017.

[19] Shu-Wei Lu, Yi-Hsuan Tsai, and Yi-Ting Chen. Toward Real-World BEV Perception: Depth Uncertainty Estimation via Gaussian Splatting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 17124–17133, 2025.

[20] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.

[21] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023.

[22] Ethan Perez, Florian Strub, Harm De Vries, Vincent Dumoulin, and Aaron Courville. Film: Visual reasoning with a general conditioning layer. In *Proceedings of the AAAI conference on artificial intelligence*, volume 32, 2018.

[23] Jonah Philion and Sanja Fidler. Lift, splat, shoot: Encoding images from arbitrary camera rigs by implicitly unprojecting to 3d. In *European conference on computer vision (ECCV)*, pages 194–210, 2020.

[24] Hossein Pirayesh and Huacheng Zeng. Jamming attacks and anti-jamming strategies in wireless networks: A comprehensive survey. *IEEE communications surveys & tutorials*, 24(2):767–809, 2022.

[25] Lennart Reiher, Bastian Lampe, and Lutz Eckstein. A Sim2Real Deep Learning Approach for the Transformation of Images from Multiple Vehicle-Mounted Cameras to a Semantically Segmented Image in Bird's Eye View. In *IEEE Intelligent Transportation Systems Conference (ITSC)*, pages 1–7, 2020.

[26] Yujiao Shi and Hongdong Li. Beyond Cross-View Image Retrieval: Highly Accurate Vehicle Localization Using Satellite Image. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 17010–17020, 2022.

[27] Yujiao Shi, Fei Wu, Akhil Perincherry, Ankit Vora, and Hongdong Li. Boosting 3-dof ground-to-satellite camera localization accuracy via geometry-guided cross-view transformer. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 21516–21526, 2023.

[28] Zhenbo Song, Jianfeng Lu, Yujiao Shi, et al. Learning Dense Flow Field for Highly-Accurate Cross-View Camera Localization. *Advances in Neural Information Processing Systems (NeurIPS)*, 36:70612–70625, 2023.

[29] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems (NeurIPS)*, 30, 2017.

[30] Shan Wang, Yanhao Zhang, Akhil Perincherry, Ankit Vora, and Hongdong Li. View Consistent Purification for Accurate Cross-View Localization. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 8197–8206, 2023.

[31] Shan Wang, Chuong Nguyen, Jiawei Liu, Yanhao Zhang, Sundaram Muthu, Fahira Afzal Maken, Kaihao Zhang, and Hongdong Li. View from Above: Orthogonal-View Aware Cross-View Localization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 14843–14852, 2024.

[32] Sijie Wang, Rui She, Qiyu Kang, Siqi Li, Disheng Li, Tianyu Geng, Shangshu Yu, and Wee Peng Tay. Multi-Modal Aerial-Ground Cross-View Place Recognition with Neural ODEs. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11717–11728, 2025.

[33] Xiaolong Wang, Runsen Xu, Zhuofan Cui, Zeyu Wan, and Yu Zhang. Fine-Grained Cross-View Geo-Localization Using a Correlation-Aware Homography Estimator. *Advances in Neural Information Processing Systems (NeurIPS)*, 36:5301–5319, 2023.

[34] Yue Wang, Vitor Campagnolo Guizilini, Tianyuan Zhang, Yilun Wang, Hang Zhao, and Justin Solomon. DETR3D: 3D Object Detection from Multi-View Images via 3D-to-2D Queries. In *Conference on Robot Learning (CoRL)*, pages 180–191, 2022.

[35] Panwang Xia, Qiong Wu, Lei Yu, Yi Liu, Mingtao Xiong, Lei Liang, Yongjun Zhang, and Yi Wan. Revisiting Cross-View Localization from Image Matching. *arXiv preprint arXiv:2508.10716*, 2025.

[36] Zimin Xia and Alexandre Alahi. FG^2: Fine-Grained Cross-View Localization by Fine-Grained Feature Matching. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6362–6372, 2025.

[37] Zimin Xia, Olaf Booij, and Julian F. P. Kooij. Convolutional Cross-View Pose Estimation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(5): 3813–3831, 2023.

[38] Zimin Xia, Chenghao Xu, and Alexandre Alahi. Loc^2: Interpretable Cross-View Localization via Depth-Lifted Local Feature Matching. In *International Conference on Learning Representations (ICLR)*, 2025.

[39] Xia Yuan, Kaiyang Wang, Riyu Qin, and Jiachen Xu. A Multi-Modality Ground-to-Air Cross-View Pose Estimation Dataset for Field Robots. *Scientific Data*, 12(1):754, 2025.

[40] Tianyi Zhang and Matthew Johnson-Roberson. Learning cross-scale visual representations for real-time image geo-localization. *IEEE Robotics and Automation Letters*, 7 (2):5087–5094, 2022.

[41] Hongyu Zhou, Zheng Ge, Zeming Li, and Xiangyu Zhang. MatrixVT: Efficient Multi-Camera to BEV Transformation for 3D Perception. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

[42] Sijie Zhu, Taojiannan Yang, and Chen Chen. Vigor: Cross-view image geo-localization beyond one-to-one retrieval. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3640–3649, 2021.

[43] Sijie Zhu, Mubarak Shah, and Chen Chen. TransGeo: Transformer Is All You Need for Cross-View Image Geo-Localization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1162–1171, 2022.

[44] Xizhou Zhu, Weijie Su, Lewei Lu, Bin Li, Xiaogang Wang, and Jifeng Dai. Deformable detr: Deformable transformers for end-to-end object detection. *arXiv preprint arXiv:2010.04159*, 2020.

[45] Jasmine Zidan, Elijah I Adegoke, Erik Kampert, Stewart A Birrell, Col R Ford, and Matthew D Higgins. Gnss vulnerabilities and existing solutions: A review of the literature. *IEEE access*, 9:153960–153976, 2020.